

An Efficient Collaborative Task Offloading Approach Based on Multi-Objective Algorithm in MEC-Assisted Vehicular Networks

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Abstract—Mobile edge computing assisting vehicular networks to achieve collaborative computing is a significant research area, as it can effectively expand the on-board computational resources, thereby improving the responsiveness of vehicular applications and reducing energy consumption. However, the existing task offloading schemes still have various gaps and face challenges that should be addressed because task offloading problems are usually multi-objective optimization problems (MOOP) and NP-hard problems. In this respect, we formulate a new multi-objective model for vehicle-centered task offloading taking into account the queuing and computing processes of tasks. The minimization of the average delay time, average energy consumption, and average payment cost for each vehicle is defined as a MOOP. We propose an efficient collaborative offloading approach based on task triage strategy and multi-objective optimization to address the model. Specifically, this approach can be decomposed into two stages. First, a task triage strategy is developed based on a comprehensive analysis of task urgency and task criticality to make offloading decisions for triaged tasks with extreme characteristics. Then, we propose a novel multi-objective optimization framework for the MOOP. Finally, we develop a multi-objective hybrid genetic algorithm that integrates the associative learning immediate memory strategy and genetic operations to solve the MOOP. Extensive experiments and performance evaluations based on real-world traces of taxis demonstrate the effectiveness of the proposed approach. The results indicate that the approach outperforms six other well-known multi-objective algorithms in solving the MOOP.

Index Terms—Mobile edge computing, task offloading, multi-objective optimization, task urgency, task criticality, Bayesian maximum entropy.

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I. INTRODUCTION

AS CONNECTED autonomous vehicles (CAVs) integrate more and more advanced vehicular applications, such as AR navigation, natural language processing and assisted driving, the connectivity and intelligence of the vehicles undergo a noticeable and continual enhancement [1], [2]. Meanwhile, the limited computational resources and storage capabilities of individual vehicles have become bottlenecks for these applications, which naturally leads to high energy consumption and latency [3]. Nowadays, vehicular networks assisted by mobile edge computing (MEC) have been recognized as one of the most promising advancements in the transportation domain [4]. MEC, as a novel distributed computing paradigm, supports tasks with low latency requirements in vehicular networks by offloading resources at the network edge [4], [5]. However, due to the increasing number of vehicles and dynamic distribution of vehicle density, as well as the heterogeneous computation capacities of MEC servers, it is essential for the network operator to design intelligent offloading strategies to improve network performance and provide high-quality services for users. Therefore, how to make offloading decision for CAVs has become a critical issue.

Although great efforts have been focused on task offloading in MEC-assisted vehicular networks, the existing task offloading schemes still have various gaps and primarily face the following issues. Firstly, task offloading problems are usually multi-objective optimization problems (MOOP) and NP-hard problems [6], [7]. In terms of task offloading strategies, current research mainly adopts multi-objective optimization (MOO) methods, such as the improved NSGA-II and NSGA-III methods, to obtain a diverse and high-quality solution set and guides offloading decisions [8], [9]. However, for some large-scale, high-dimensional and complex practical problems, there are some remaining issues such as premature convergence, low optimization accuracy and susceptibility to local optimization. Secondly, for the characterization of vehicular task attributes, existing work mainly focuses on describing the input data size, computational load requirements and execution time constraints of the task [2], [6], while ignoring the realistic characteristics of vehicle applications, such as the criticality of the task. In general, for CAVs, the vehicular systems are mostly typical mixed criticality systems, and each vehicular application needs to share limited computing and storage resources on a shared computing

platform [10]. When the demand for high-criticality applications increases or an emergency occurs, the system may preferentially allocate resources to these tasks, leading to low-criticality tasks being preempted and missing their deadlines [11]. Differentiated task offloading strategies must be adopted for these two applications due to their varying criticality levels and Quality of Service (QoS) requirements. Additionally, many existing task offloading schemes focus on offloading tasks to MEC-enabled roadside units (MRSUs) through the wireless access network vehicular-to-roadside, thereby optimizing energy consumption and task execution time of edge systems [2], [5]. To date, most work on this topic have focused on optimizing energy, time and other metrics from the perspective of network operators, while ignoring the interests of vehicle users. Notably, existing studies usually assume that different applications, regardless of their origin, can freely exploit the three layers of cloud-edge-end computing resources. This assumption is, however, too optimistic to be implemented practically, because vehicles are most likely to be self-interested and reluctant to share their resources for security, privacy, and economic reasons, which may violate the scheduling policy made by the network operator. Therefore, it is essential to design an appropriate payoff strategy to stimulate vehicles to follow the rules made by the network operator.

Our work fills the gap of using MOO for task offloading in MEC-assisted vehicular networks and adopts, in an innovative way, a two-stage collaborative offloading approach to obtain the optimal offloading decision. A task triage strategy is constructed in the first stage, which is based on the comprehensive analysis of task urgency and task criticality. The task triage strategy can effectively identify a subset of tasks with extreme characteristics and make offloading decisions for them. The remaining tasks after completing the first stage are used as input in the second stage of the proposed method. In the second stage, we formulate the vehicle-centric multi-objective task offloading problem by modeling the procedure of the remaining task queuing, task upload and computation based on queuing theory, which aims at minimizing the average delay time, energy consumption, and payment cost for each vehicle. To tackle the multi-objective problem, a novel optimization framework based on Bayesian maximum entropy (BME) is developed. The proposed framework integrates the associative learning immediate memory (ALIM) strategy and genetic algorithm to develop a new multi-objective hybrid genetic algorithm (MOHGA) for solving the problem at hand. This two-stage approach utilizes a combination of precise allocation and approximate solution, which not only reduces the task scale, improves computational efficiency, but also takes into account the global optimization objective. The contributions are summarized as follows:

- 1) A new vehicle-centric multi-objective task offloading (MOTO) model is proposed in this paper, which aims at minimizing the average delay time, average energy consumption, and average payment cost for each CAV. A queuing theory approach is adopted by the model to construct the average waiting time of task conditions during local processing and offloading.

- 2) Considering the significant impact of different types of task characteristics on the offloading process, a task offloading strategy has been implemented based on a comprehensive analysis of task urgency and task criticality to make offloading decisions for triaged tasks with extreme characteristics.
- 3) A BME-based multi-objective optimization framework (MOOF) is developed. Based on the framework, we proposed a novel multi-objective algorithm, namely MOHGA, which addresses the MOTO by incorporating an ALIM strategy and genetic operations.
- 4) Several experiments have been conducted on instances of different scales based on real-world traffic data in Wuhan, China, to verify the effectiveness of the designed models and algorithms. The extensive experimental results show that they perform better in solving the MOTO than their peers.

The rest of the paper is organized as follows. Section II discusses the main related works. Section III presents the system model and problem formulation. Section IV details how to develop the BME as a new MOOF for MOTO and the algorithm design is presented in Section V. The simulations experiments and the related performance evaluations are given in Section VI. Finally, Section VII concludes this article also discussing possible future research directions.

II. RELATED WORK

In this section, we review the existing work on task offloading problems that are often MOOP and attracts many researchers to study in this field. Researchers have proposed multiple offloading mechanisms based on MOO, mainly including methods based on multi-objective evolutionary algorithms and multi-objective reinforcement learning. Some works have adopted a MOO method based on genetic algorithms to address this issue. For instance, Gao et al. [8] and Cui et al. [12] both improved the non-dominated sorting genetic algorithm (NSGA-II) to find the optimal solution for the constrained MOOP of offloading task delay and energy consumption. Similarly, Li et al. [13] improved evolutionary algorithm based on NSGA-III and applied it to the optimization problems of latency, cost, energy consumption, load balancing, and user satisfaction during computation offloading. In addition to multi-objective genetic algorithms, there are also other computational offloading solutions based on MOO. Xu et al. [14] proposed a hybrid computation offloading framework with both MABSs and RSUs in Internet of Vehicles (IoV). They utilized the Multi-Objective Evolutionary Algorithm based on Decomposition (MOEA/D) to devise effective strategies within this framework. In [15], Wu et al. presented an improved strength Pareto-based evolutionary algorithm II (SPEA-II) based on the TOPSIS and MCDM methods to solve the vehicle task offloading problem, aiming to reduce total time and keep the load balance. In addition, some MOO mechanisms based on reinforcement learning methods have been developed to address task offloading problems. Zhang et al. [2] proposed RMDDQN-Learning, a multi-objective Double

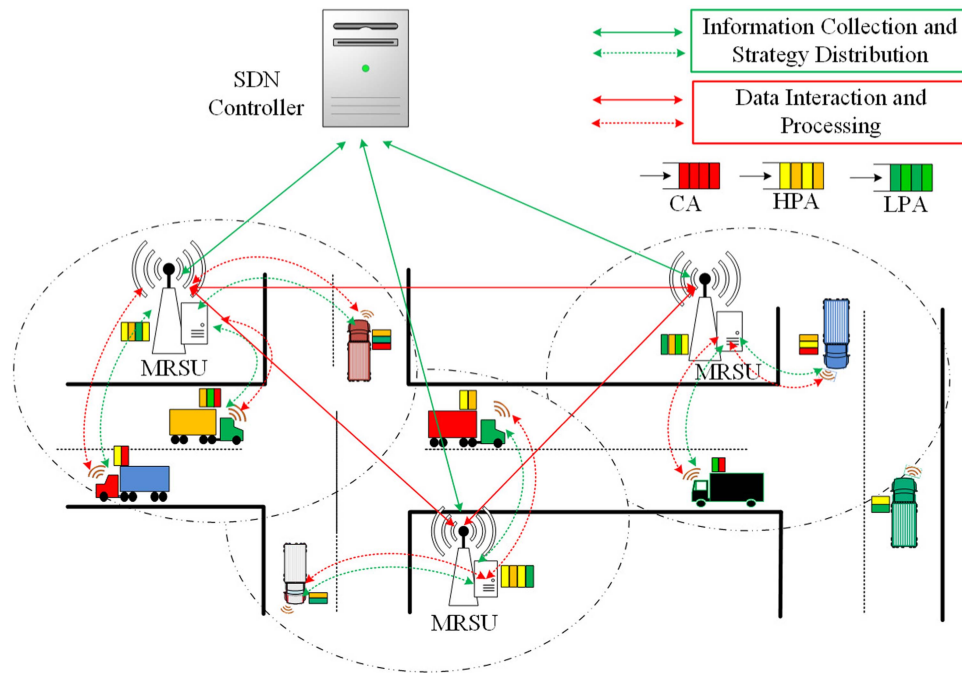


Fig. 1. The software-defined vehicular network system architecture.

TABLE I
CRITICALITY LEVEL OF VEHICLE APPLICATIONS [16], [17], [18]

Types		Applications	Offload
Critical application	CAs	Stability control, Signal violation warning, and Collision warning	No
Low-critical applications	HPAs	Navigation, Location sight recognition, High map	Yes
	LPAs	Multimedia, Video processing, Speech recognition	Yes

Deep Q-Network deep reinforcement learning algorithm for computation offloading in the IoV. This algorithm optimizes delay, energy consumption, computing load, and privacy entropy of vehicle terminals to achieve optimal offloading decisions.

However, there are still many shortcomings in current research on task offloading. On the one hand, existing solutions often ignore the characteristics of vehicular applications, resulting in inefficient resource utilization and a decrease in reliability. According to the application type, computing requirement, and delay requirement of the vehicular computing tasks, the vehicular computing tasks are mainly categorized into critical application (CA), high-priority application (HPA) and low-priority application (LPA) [16], [17], as shown in Table I. CAs are safety-related tasks with the highest priority, including adaptive cruise control, stability control, and automatic braking, etc. Because of the extremely high requirements for service reliability and latency, it is generally executed by the vehicle itself and is not permitted to be offloaded to edge servers or other vehicles [17]. Therefore, automakers and tier-one suppliers should reserve sufficient computing resources for these tasks, which guarantee their reliable operation at all times. Regarding low-criticality

LPA or HPA applications like navigation and multimedia, they can be offloaded to edge servers because they have relatively soft tolerable response times and occasional failures are acceptable [17], [18]. Furthermore, vehicle systems often prioritize high-criticality applications over those of low criticality. Low-criticality applications are typically processed locally on the vehicle's on-board unit only when computing resources are sufficient. However, this practice ignores applications' urgency to get the service, and thus, results in a great sacrifice for low-critical applications [18]. The emergence of MEC enables vehicles to use external edge servers to improve the QoS of tasks. In view of this, to effectively manage these vehicular computing tasks with different attributes and provide high-quality services, it is necessary to design a more comprehensive task model and an effective task offloading strategy. On the other hand, with the increasing number of vehicles and computing tasks, effective offloading solutions have become an urgent need to improve the performance of on-board systems. The MOTO problem is difficult to solve with conventional MOO methods and algorithms because there are not only conflicts among multiple optimization objectives, but also the search space enlarges exponentially as the number of decision variables increases. Therefore, it is necessary to design a new multi-objective method to better address the challenges of MOTO problems and guarantee the convergence efficiency and distributivity at the same time.

Unlike the above research work, we propose an efficient collaborative offloading approach based on task triage strategy and MOO to address task offloading problem. Firstly, a task triage strategy is implemented to triage tasks based on a comprehensive analysis of task urgency and criticality. Secondly, the offloading decisions of the remaining tasks are optimized based on the

TABLE II
KEY NOTATIONS AND DESCRIPTIONS

Notation	Description
V	set of CAVs
C	set of MRSUs
T_v	set of tasks for CAV v
N	number of tasks per CAV v
v	CAV index $v \in V$
n	CAV task index $n, n \in N$
c	MRSU index $c, c \in C$
w_{vn}	required computation resource of CAV v 's task
ω_{vn}	computation data size of CAV v 's task
τ_{vn}	the time limit of the computation task
cr_{vn}	the critical attribute of task n
p_v	the proportion of offloaded tasks
p_v^N	the normalized workload of the CAV v
p_v^T	transmit powers of CAV v
c_v^C	computation capability of CAV v
λ_v	task arrival rate of CAV v
η_v	computational density of CAV v 's task
C_E	computation capability of MRSU
π_v^E, π_v^b	the charging prices for unit resources on the MRSU and for each bit transmission task, respectively.
rew_{vn}	the reward after each task offloading
δ_U, δ_L	the upper and lower truncation points of the task urgency data, respectively.
OD_{vn}	the offloading decision of n -task of v -th CAV

novel MOO method. The specific model-building process will be described in the following section.

III. SYSTEM MODEL

In this section, we consider a software-defined vehicular network architecture for task offloading, which consists of vehicles, MRSUs, and a software-defined network (SDN) controller, as shown in Fig. 1. Our two-stage collaborative offloading algorithm is deployed on the SDN controller, which serves as an orchestrator for information collection, strategy distribution, and infrastructure collaboration. The SDN controller and MRSU are directly connected via a fiber-optic communication link. The MRSU is then connected to the vehicles through V2X, which makes the vehicles indirectly connected to the SDN controller via the MRSU, thus forming an 'SDN controller-MRSU-vehicle' model [5], [19]. The key feature of these network entities is to decouple data transmission and processing from control, and adhere to the OpenFlow protocol [19]. Specifically, When the vehicles move within an RSU's coverage, they can access the MRSU via a wireless channel to request edge assistance from the SDN controller [20]. The SDN controller receives vehicle task requests and related information from the MRSU, analyzes both the current network status and the computing resource status of the MRSU, formulates the optimal task offloading and data caching strategy, and then sends the decision results to the MRSUs. Finally, the vehicles and the MRSUs execute the corresponding operations based on the instructions from the SDN controller to complete task offloading and data processing [19], [20]. The tasks in the CAVs can be executed locally or offloaded to the MEC servers. Similar to [2], [6], [21], considering the tiny size of the task result, we neglect the task transmission delay of the task result. For easy reference, Table II summarizes the key notations and their definitions used in this article.

A. Preliminary

We consider there are c MRSUs, denoted as a set $C = \{1, 2, \dots, c\}$, deployed near a crossroad and connected by a high-speed LAN. There are ν vehicles arriving at the crossroad area, denoted as a set $V = \{1, 2, \dots, \nu\}$, which are overlaid by the MRSUs nearby. Each CAV v is modeled as a 2-tuple (c_v^C, p_v^T) , where $v \in V$, c_v^C represents the computing capability of v , and p_v^T is the transmission power to offload tasks. The computation tasks generated by the application of each CAV v are assumed to follow a Poisson process with an average arrival rate λ_v . This assumption is plausible and has been widely adopted in many works [1], [21]. The tasks are independent of each other and cannot be subdivided. For each CAV v , an array of applications is executed, which accordingly generate a series of computation tasks, and a set of tasks is denoted by $T_v = \{t_{v1}, t_{v2}, \dots, t_{vn}, \dots, t_{vN}\}$. Each task t_{vn} can be represented by a tuple with four parameters: $t_{vn} = \langle \omega_{vn}, w_{vn}, \tau_{vn}, cr_{vn} \rangle$, where ω_{vn} represents the data size (in terms of bits) of the task. $w_{vn} = \omega_{vn}\eta_v$ represents the computation amount (in terms of CPU cycles) needed by the task when processed on the CAV or on a MRSU, and η_v (in terms of CPU cycles/bit) is the computational density of CAV v ' task, τ_{vn} indicates the tolerable latency for task t_{vn} , and cr_{vn} represents the criticality level attribute of the task. We define cr_{High} and cr_{Low} to represent HPAs and LPAs respectively, and denoted as $cr_{vn} \in (cr_{High}, cr_{Low})$. The set $OD_{vn} = \{0, 1\}$ represents the offloading decision for each task. Each task can either be executed locally ($OD_{vn} = 0$) or offloaded to MRSU via V2X ($OD_{vn} = 1$).

B. Task Processed Locally

In this paper, we consider the task computation process at the CAV as an M/M/1 queue [4]. The CAV v chooses to offload the task with a probability p_v ($0 \leq p_v \leq 1$). The value of p_v is calculated as the ratio of the number of offloaded tasks to the number of all generated tasks by CAV v i.e., $p_v = \sum_{n=1}^N OD_{vn}/N$. Accordingly, the average rate of tasks offloaded to the edge follows a Poisson process with an average rate of $p_v\lambda_v$, while the tasks processed locally also follow a Poisson process with an average rate of $(1 - p_v)\lambda_v$. According to the M/M/1 model [21], the time of the task spent on the vehicle system is the sum of the computation time of the task and the average waiting time for execution. It can be calculated as follows:

$$\tau_{vn}^L = \frac{w_{vn}(1 - OD_{vn})}{c_v^C(1 - p_v^N)} + \frac{(1 - p_v)\lambda_v}{\mu_v^V[\mu_v^V - (1 - p_v)\lambda_v]}. \quad (1)$$

where μ_v^V is the average execution rate of the locally processed tasks, i.e., $\mu_v^V = c_v^C(1 - p_v^N)/\bar{w}_v$. $\bar{w}_v$ represents the average number of CPU cycles required for local execution of CAV v , which can be obtained as follows: $\bar{w}_v = \sum_{n=1}^N (1 - OD_{vn})w_{vn}/\sum_{n=1}^N (1 - OD_{vn})$. In addition, we define $\rho_v^L = (1 - p_v)\lambda_v/\mu_v^V$ as the service intensity, which indicates the busyness of CAV v local processing tasks. Then the energy consumption of the local computing task of CAV v can be expressed as:

$$E_{vn}^L = \kappa_v(1 - OD_{vn})[c_v^C(1 - p_v^N)]^2 w_{vn}. \quad (2)$$

where κ_v is an energy coefficient of the CAV v [20].

C. Task Uploading Model

The transmission time of offloading data from CAV v to MRSU is as follows:

$$\tau_{vn}^U = \frac{\omega_{vn} \cdot OD_{vn}}{R_{vn}^V} + \frac{p_v \lambda_v}{\mu_v^U (\mu_v^U - p_v \lambda_v)}. \quad (3)$$

where μ_v^U is the average execution rate of task offloading in the vehicle system, i.e., $\mu_v^U = R_v / \bar{\omega}_v$. $\bar{\omega}_v$ is the average data size of offloaded tasks of CAV v , which can be calculated as follows: $\bar{\omega}_v = \sum_{n=1}^N OD_{vn} \omega_{vn} / \sum_{n=1}^N OD_{vn}$. R_{vn}^V is the task uploading transmission rate from v to MRSU, which can be obtained according to the literature [21]: $R_{vn}^V = B_v \log(1 + p_v^T g_{vn} d_{vn}^{-pl} / \delta^2)$, where g_{vn} and B_v represent the channel gain and bandwidth between CAV v and MRSU, respectively. p_v^T represents the power of the CAV to send the task to its MRSU. δ^2 represents the noise power parameter, d_{vn} is the distance between CAV v and MRSU, and pl is the path loss exponent. In addition, we define $\rho_v^U = p_v \lambda_v / \tau_v^U$ to represent the service intensity of CAV v transmission task. And the energy consumption for transmitting the tasks from the CAV to the MRSU can be obtained as follows: $E_{vn}^U = p_v^T \tau_{vn}^U$.

D. MRSU Computation Model

We define the SDN controller-MRSU mode with c homogeneous edge servers, modeled as an M/M/c queue. Consequently, the arrival rate of tasks is equal to the sum of all arrival rates of offloaded tasks from multiple CAVs, i.e., $\lambda_{MEC} = \sum_{v=1}^V p_v \lambda_v$. Based on the M/M/c queuing model and Erlang's formula [26], the time of tasks spent on the MRSU can be calculated as follows:

$$\tau_{vn}^E = \frac{OD_{vn} \omega_{vn}}{C_E} + \frac{C(c, \rho_E)}{\mu_v^E \cdot c - \lambda_{MEC}} \quad (4)$$

where $\mu_v^E = C_E / \bar{e}_v$ represents the average execution rate of each edge server. $\bar{e}_v = \sum_{n=1}^N OD_{vn} \omega_{vn} / \sum_{n=1}^N OD_{vn}$ represents the average number of CPU cycles required for offloading tasks. $\rho_E = \lambda_{MEC} / \mu_v^E c$ represents the service intensity of the c th edge server. The calculation formula of $C(c, \rho_E)$ is as follows [22]:

$$C(c, \rho_E) = \frac{\frac{(c\rho_E)^c}{c!} \left(\frac{1}{1-\rho_E} \right)}{\sum_{u=0}^{c-1} \frac{(c\rho_E)^u}{u!} + \frac{(c\rho_E)^c}{c!} \left(\frac{1}{1-\rho_E} \right)}. \quad (5)$$

E. Vehicle Payment Cost

The offloading costs for CAVs mainly consist of service fees charged by operators, such as MEC server resource usage fees and task transmission fees. In general, CAVs tend to exhibit rational behavior, which may prioritize their own interests, potentially conflicting with the optimal offloading strategies established by the network operator [1]. To facilitate this, it is essential for operators to guide CAV offloading decisions by setting incentive fees to adjust the trade-off between service fees and vehicle offloading costs. Therefore, the offloading cost for

each task of CAV can be computed as follows:

$$C_v^{ave} = \pi_v^E \sum_{n=1}^N OD_{vn} \omega_{vn} + \pi_v^b \sum_{n=1}^N OD_{vn} \omega_{vn} - \sum_{n=1}^N OD_{vn} rew_{vn}. \quad (6)$$

where π_v^E represents the payment for resources on the MRSU server, i.e., the price per CPU cycle. π_v^b represents the price per bit of task transmission and rew_{vn} represents the reward obtained by a CAV after completing an offloading task.

F. Problem Formulation

Based on the model established in the previous section, we can use T^{ave} , E^{ave} and C^{ave} to represent the average delay time, average energy consumption and the average cost of each CAV, respectively.

$$T^{ave} = \frac{1}{V} \sum_{v=1}^V \sum_{n=1}^N (\tau_{vn}^L + \tau_{vn}^U + \tau_{vn}^E) \quad (7)$$

$$E^{ave} = \frac{1}{V} \sum_{v=1}^V \sum_{n=1}^N (E_{vn}^L + E_{vn}^S). \quad (8)$$

$$C^{ave} = \frac{1}{V} \sum_{v=1}^V C_v^{ave}. \quad (9)$$

This paper optimizes multiple indicators involved in the execution of CAV tasks, so as to keep the average delay time, average energy consumption and average payment cost of the entire computation offloading process to a minimum. The optimization problem can be formulated as:

$$\begin{aligned} \text{P: } \min \{ & E^{ave}, T^{ave}, C^{ave} \} \\ \text{s.t. } & C1 : (1 - p_v) \lambda_v < \mu_v^V \\ & C2 : p_v \lambda_v < \mu_v^U \\ & C3 : \lambda_{MEC} < \mu_v^E c \\ & C4 : \rho_v^L < 1 \& \rho_v^U < 1 \& \rho_E < 1 \\ & C5 : OD_{vn} = \{0, 1\} \\ & C6 : 0 < p_v^N < 1 \\ & C7 : \max \{ \tau_{vn}^L, \tau_{vn}^U + \tau_{vn}^E \} \leq \tau_{vn}. \end{aligned} \quad (10)$$

$C1$ guarantees that the task arrival rate of local execution should not exceed the CAV's processing rate. $C2$ defines that the data workload of offloaded tasks is less than the average execution rate. $C3$ enforces that the workload arrival rates should not exceed the service rate of the MEC. $C4$ ensures that the task queuing system is in a steady-state. $C5$ shows that the offloading decision is binary. $C6$ ensures that the CAV has sufficient computing resources without exceeding its CPU capacity. $C7$ indicates that the completion time of task n of CAV v cannot exceed its tolerable latency.

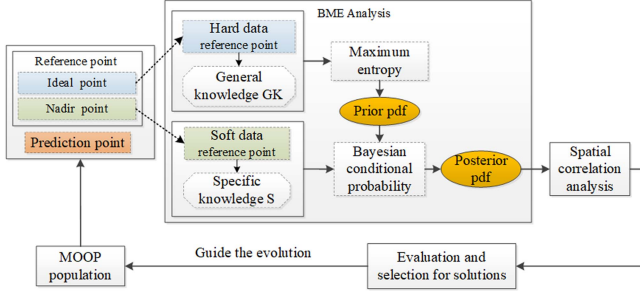


Fig. 2. Main concept of BME-based MOOF.

IV. PROPOSED MOOF

In this section, we first introduce some basics about BME, then we propose a novel MOOF based on BME to solve the MOOP of MOTO. The main concept of BME-based MOOF is shown in Fig. 2. It integrates a reference point strategy with evolutionary algorithm search operations to achieve the complementary and collaborative utilization of both ideal and nadir reference points. The core idea is to map solutions from the high-dimensional objective space to a spatial random field. By using both hard data and soft data reference sequences, it effectively addresses MOOP with diverse Pareto frontiers by inferring and evaluating the positions of solutions in the random field.

A. Basic BME

The principle of BME involves estimating the value z_k at location x_k , by employing h hard data points z_{hard} , represented as $z_{\text{hard}} = [z_1, \dots, z_h]$, and incorporating s soft data points z_{soft} , denoted as $z_{\text{soft}} = [z_1, \dots, z_s]$, along with generalized knowledge (GK) derived from multiple data sources [23]. Hard data is defined as information with minimal or negligible errors, thereby offering a high degree of accuracy. Conversely, soft data refers to information characterized by a certain degree of uncertainty and is commonly presented in formats such as intervals or expressions based on probability [24]. The soft data in this paper is presented as intervals, where the lower and upper limits of the interval are denoted by 'lp' and 'up', respectively, as illustrated in (11) [25].

$$z_{\text{soft}} = \{[z_1, \dots, z_s]\}, z_s \in [lp_s, up_s]. \quad (11)$$

To estimate the value at point x_k within the BME framework, it is necessary to utilize an information entropy-based measure to quantify the information contained in the prior probability density function (pdf) $f_{GK}(z_{\text{all}})$. GK serves as a mathematical constraint to maximize the information entropy. $f_{GK}(z_{\text{all}})$ represents the prior pdf of the multidimensional random variable $z_{\text{all}} = [z_{\text{hard}}, z_{\text{soft}}, z_k]$. In this paper, GK is defined as the covariance function derived from the statistical analysis of known data, which includes both hard and soft data sets. Subsequently, in accordance with Bayes' theorem, the prior pdf conditional upon the soft data is transformed into the posterior probability distribution, as illustrated in (12) [25].

$$f_k(z_k) = A_1^{-1} \int_{lp}^{up} f_{GK}(z_{\text{all}}) dz_{\text{soft}}. \quad (12)$$

where, $A_1 = \int_{lp}^{up} f_{GK}(z_{\text{data}}) dz_{\text{soft}}$ is the normalization constant, $f_{GK}(z_{\text{all}}) = \int f_{GK}(z_{\text{data}}) dz_k$, $f_{GK}(z_{\text{data}})$ represents the joint pdf of both hard data and soft data. $z_{\text{data}} = [z_{\text{hard}}, z_{\text{soft}}]$, which typically represents specialized knowledge (SK).

Finally, the mean of the posterior pdf can be computed and utilized as the estimation for variable z_k at location x_k . Clearly, a crucial aspect of implementing the BME method for multi-objective analysis requires the systematic collection and organization of all feasible hard and soft data. However, the BME method cannot be directly applied to MOOP and necessitates appropriate extensions.

B. Hard Data and Soft Data

The reference point data consists of a hard data reference matrix (denoted as R_h) and a soft data reference matrix (denoted as R_s). The detailed construction steps for both matrices are presented as follows:

- 1) Divide the initial population P_0 into M equal-sized sub-populations, where M is the number of objectives, G is the number of optimizations for each subpopulation.
- 2) Parallel optimize the i th subpopulation to obtain the minimum value and the maximum value of the i th objective by a single objective algorithm. These values are then designated as the ideal point $f_{hi}^{\min}(x)$ and the nadir point $f_{si}^{\max}(x)$ for the i th objective, respectively, where $i \in \{1, 2, \dots, G\}$ and $j \in \{1, 2, \dots, M\}$. All M objectives are optimized simultaneously in parallel.
- 3) Apply Min-Max Normalization to the ideal points and the nadir points of the M sub-objectives from all optimization generations, subsequently forming the R_h and R_s , respectively. Both R_h and R_s are matrices of size $G \times M$, as shown in (13) and (14) respectively.

$$R_h = [p_{h1}; p_{h2}; \dots; p_{hi}; \dots; p_{hG}]. \quad (13)$$

$$R_s = [p_{s1}; p_{s2}; \dots; p_{si}; \dots; p_{sG}]. \quad (14)$$

where, $p_{hi} = [f_{hi1}^{\min}, f_{hi2}^{\min}, \dots, f_{hij}^{\min}, \dots, f_{hiM}^{\min}]$; $p_{si} = [f_{si1}^{\max}, f_{si2}^{\max}, \dots, f_{sij}^{\max}, \dots, f_{siM}^{\max}]$, $i \in \{1, 2, \dots, G\}$, $j \in \{1, 2, \dots, M\}$.

C. Spatial Simulation and Mapping

In spatial simulation, the initial step requires combining R_h and R_s to form the global matrix $R_0 = [R_h, R_s]$ for Euclidean distance measurement. Subsequently, the covariance matrix C_R of R_0 is calculated using an exponential covariance model based on Euclidean distance. The exponential covariance model is presented below [26].

$$\text{cov}(p_i, p_j) = e^{-(d_h(p_i, p_j)/\varepsilon)}. \quad (15)$$

where p_i and p_j represent any two data points in the R_0 , with d_h denoting the Euclidean distance between them. ε represents the covariance range.

We use Cholesky decomposition to perform a single simulation of the covariance matrix C_R , to generate simulated values z_h of R_h and simulated values z_s of R_s that conform to the multivariate normal distribution. Consequently, we establish the

hard data reference sequence $\mathbf{X}_h$ for $\mathbf{R}_h$ and the soft data reference sequence $\mathbf{X}_s$ for $\mathbf{R}_s$ to describe the spatial mapping relationship between multi-objective values and simulated values. The two sequences are as follows, respectively.

$$\mathbf{X}_h = \{[\mathbf{p}_{h1}, z_{h1}]; [\mathbf{p}_{h2}, z_{h2}]; \dots; [\mathbf{p}_{hG}, z_{hG}]\}. \quad (16)$$

$$\mathbf{X}_s = \{[\mathbf{p}_{s1}, z_{s1}]; [\mathbf{p}_{s2}, z_{s2}]; \dots; [\mathbf{p}_{sG}, z_{sG}]\}. \quad (17)$$

D. BME Estimates

In this paper, the mean $\bar{\mathbf{z}}$ and function $\mathbf{C}$ of the GK are computed from $\mathbf{X}_h$ and $\mathbf{X}_s$, thereby obtaining the prior probability density function $f_{\text{GK}}(\mathbf{z}_{\text{all}})$, which takes the form of a multivariate Gaussian distribution [27], as depicted in (18).

$$\Phi(\mathbf{z}; \bar{\mathbf{z}}; \mathbf{C}) = 2\pi^{-G/2} |\mathbf{C}|^{-1/2} \times \exp \left[-(\mathbf{z} - \bar{\mathbf{z}})^T \mathbf{C}^{-1} (\mathbf{z} - \bar{\mathbf{z}}) / 2 \right]. \quad (18)$$

where $\mathbf{C}$ represents the covariance matrix for $\mathbf{z}_{\text{all}} = [\mathbf{X}_h, \mathbf{X}_s, \mathbf{X}_k]$, $\mathbf{X}_k$ represents the evaluation sequence between the yet-to-be-evaluated multi-objective values and simulated values in the new population, and can be expressed as:

$$\mathbf{X}_k = \{[\mathbf{p}_{k1}, z_{k1}]; [\mathbf{p}_{k2}, z_{k2}]; \dots; [\mathbf{p}_{kG_{\text{max}}}, z_{kG_{\text{max}}}] \}. \quad (19)$$

where, $\mathbf{p}_{kl} = [f_{kl1}, f_{kl2}, \dots, f_{klj}, \dots, f_{klM}]$ is the matrix of multi-objective function values obtained in the i th iteration for the new population, and z_{kl} is the simulated value of $\mathbf{p}_{kl}$ to be evaluated, $l \in \{1, 2, \dots, G_{\text{max}}\}$, $j \in \{1, 2, \dots, M\}$. Similar to (13) and (14), the evaluation matrix $\mathbf{R}_k$ of $\mathbf{X}_k$, represented as $\mathbf{R}_k = [\mathbf{p}_{k1}; \mathbf{p}_{k2}; \dots; \mathbf{p}_{kG_{\text{max}}}]$, is formed by aggregating the multi-objective function values $\mathbf{p}_{kl}$ over G_{max} iterations.

The partition matrix of matrix $\mathbf{C}$ is defined based on the characteristics of the multivariate Gaussian probability density function, which can be described as follows [25]:

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{hs,hs} & \mathbf{C}_{hs,k} \\ \mathbf{C}_{k,hs} & \mathbf{C}_{k,k} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_{hh} & \mathbf{C}_{hs} & \mathbf{C}_{hk} \\ \mathbf{C}_{sh} & \mathbf{C}_{ss} & \mathbf{C}_{sk} \\ \mathbf{C}_{kh} & \mathbf{C}_{ks} & \mathbf{C}_{kk} \end{bmatrix}. \quad (20)$$

where the subscripts denote covariances between various combinations of hard data-soft data-estimation points. The posterior pdf is obtained by inserting the Gaussian pdf $f_{\text{GK}}(\mathbf{z}_{\text{all}}) = \Phi(\mathbf{z}; \bar{\mathbf{z}}; \mathbf{C})$ into (12), and then using the properties of the conditional distribution of the Gaussian pdf and the law of total probability, we obtain the following formulation [27]:

$$f_k(\mathbf{z}_k) = A^{-1} \Phi(\mathbf{z}_k; \mathbf{B}_{k|h} \mathbf{z}_h; \mathbf{C}_{k|h}) \int_{\mathbf{lp} - \mathbf{B}_{s|kh} \mathbf{z}_{kh}}^{\mathbf{up} - \mathbf{B}_{s|kh} \mathbf{z}_{kh}} \Phi(\mathbf{z}_s; \mathbf{0}; \mathbf{C}_{s|kh}) d\mathbf{z}_s \quad (21)$$

where $\mathbf{B}_{k|h} = \mathbf{C}_{kh} \mathbf{C}_{hh}^{-1}$, $\mathbf{z}_k^T = [\mathbf{z}_k^T \mathbf{z}_h^T]$, $\mathbf{C}_{k|h} = \mathbf{C}_{kk} - \mathbf{B}_{k|h} \mathbf{C}_{kh}$, $\mathbf{B}_{s|kh} = \mathbf{C}_{s,kh} \mathbf{C}_{kh,kh}^{-1}$, $\mathbf{C}_{s|kh} = \mathbf{C}_{ss} - \mathbf{B}_{s|kh} \mathbf{C}_{s,kh}$.

$\mathbf{C}_{kh,s}$, $\mathbf{C}_{kh,kh} = \begin{bmatrix} \mathbf{C}_{kk} & \mathbf{C}_{kh} \\ \mathbf{C}_{hk} & \mathbf{C}_{hh} \end{bmatrix}$, and $A = \int_{\mathbf{lp}}^{\mathbf{up}} \Phi(\mathbf{z}_s; \mathbf{B}_{s|h} \mathbf{z}_h; \mathbf{C}_{s|h}) d\mathbf{z}_s$. Then, the posterior pdf $f_k(\mathbf{z}_k)$ can be used to calculate the mean of the simulated values for each of the yet-to-be-evaluated points $\mathbf{p}_{kl}$ in the prediction matrix $\mathbf{R}_k$, as shown in (22) [25].

$$\bar{\mathbf{z}}_k = A^{-1} \int_{\mathbf{lp}}^{\mathbf{up}} \mathbf{B}_{k|h} \mathbf{z}_{\text{data}} \Phi(\mathbf{z}_s; \mathbf{B}_{s|h} \mathbf{z}_h; \mathbf{C}_{s|h}) d\mathbf{z}_s. \quad (22)$$

where $\mathbf{B}_{k|h} = \mathbf{C}_{k,hs} \mathbf{C}_{hs,hs}^{-1}$, $\mathbf{z}_{\text{data}}^T = [\mathbf{z}_h^T \mathbf{z}_s^T]$, $\mathbf{B}_{s|h} = \mathbf{C}_{sh} \mathbf{C}_{hh}^{-1}$, $\mathbf{C}_{s|kh} = \mathbf{C}_{ss} - \mathbf{B}_{s|kh} \mathbf{C}_{s,kh}$.

E. Correlation Analysis

The sequences $\mathbf{X}_h$ and $\mathbf{X}_k$ provide numerical characteristics for the position of each multi-objective solution in space, and they also quantify and characterize the dependencies and variability among solution sets. The cosine distance is used as the spatial correlation coefficient to effectively capture the correlation between $\mathbf{X}_h$ and $\mathbf{X}_k$. The cosine distance can be calculated as follows:

$$C_d(\mathbf{X}_k(l), \mathbf{X}_h(i)) = \min_i \left(1 - \left(\frac{\mathbf{X}_k(l) \cdot \mathbf{X}_h(i)}{\|\mathbf{X}_k(l)\| \|\mathbf{X}_h(i)\|} \right) \right), \quad i \in \{1, 2, \dots, G\}. \quad (23)$$

The C_d value can be used as a measure of the spatial correlation between the sequences $\mathbf{X}_h$ and $\mathbf{X}_k$. A smaller C_d value indicates a higher spatial correlation between $\mathbf{p}_{kl}$ and $\mathbf{p}_{hi}$. The solution with smaller C_d values ranks higher and is prioritized for subsequent fitness evaluation operations.

V. TWO-STAGE ALGORITHM DESIGN

In this section, we propose an efficient collaborative task offloading approach that combines task triage strategy and algorithm optimization to solve MOTO, as shown in Fig. 3. This two-stage approach utilizes a combination of precise allocation and approximate solution, which not only reduces the task scale, improves computational efficiency, but also takes into account the global optimization objective.

A. Task Triage Based on Urgency and Criticality

Each task in T_v is characterized by four parameters: data size (ω_{vn}), computational load (w_{vn}), latency requirement (τ_{vn}), and criticality level (cr_{vn}). The first three parameters collectively determine the urgency of each subtask t_{vn} . Hence, the urgency of each task is determined by using the entropy-weighted TOPSIS method, which relies on data of these three parameters. However, the entropy-weighted TOPSIS method is insufficient for fully capturing the inherent characteristics of the vehicle task, specifically its criticality, as task criticality are determined by business requirements or industry standards and are not directly correlated with task urgency. To achieve efficient offloading decisions, this paper proposes a vehicle-centric two-stage task offloading approach. A task triage strategy is constructed in the first stage, which is based on the comprehensive analysis of task urgency and task criticality. The task triage strategy can

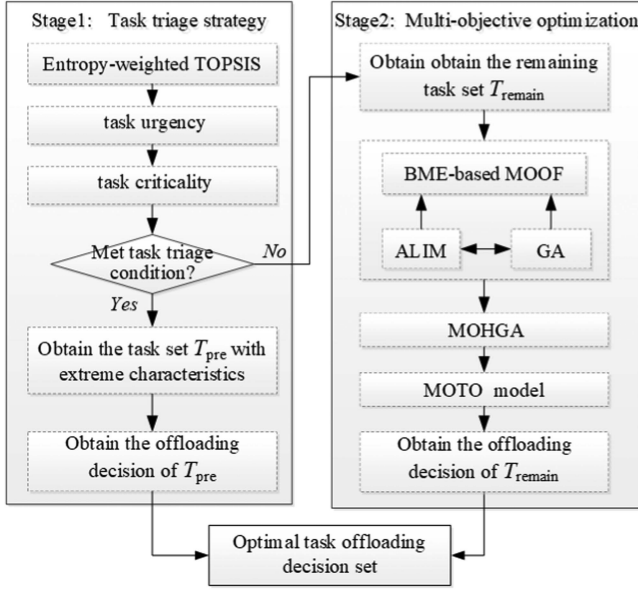


Fig. 3. Flowchart of the collaborative task offloading approach.

effectively identify a subset of tasks with extreme characteristics and make offloading decisions for them, thereby alleviating the computational burden of the multi-objective algorithm in the second stage for obtaining offloading decisions and shortening the overall computation time of the algorithm. The following section provides a comprehensive exposition. If the urgency evaluation value is large and the task criticality level is high, offloading the task to an MRSU server via a wireless link with low reliability may not be the optimal choice. Conversely, in the case of low task criticality and a small urgency evaluation value, it is advisable for vehicle systems with limited computing capacity and task competition to opt for offloading. The following rules for task triage strategies are proposed in light of this:

Definition 1: Prioritize local execution for tasks characterized by a large urgency evaluation value and high criticality level.

Definition 2: Prioritize offloading for tasks characterized by a small urgency evaluation values and low criticality levels. The specific calculation procedure for determining the weight of each task by entropy-weighted TOPSIS is given as follows [28], [29].

Step 1: Assuming that $[o_{nt}]_{N \times T}$ is the initial evaluation matrix of the TOPSIS model for evaluating task urgency, which consists of N tasks and T attributes. The task attributes differ in terms of scale and dimension and require normalization. The normalized decision matrix is shown as follows:

$$Q_0 = [e_{nt}]_{N \times T}. \quad (24)$$

where $e_{nt} = o_{nt} / \sqrt{\sum_{n=1}^N (o_{nt})^2}$, ($n = 1, 2, \dots, N; t = 1, 2, \dots, T$).

Step 2: Calculate the weighted standardized judgment matrix by using the weights based on the entropy, which can be shown as follows:

$$Q_1 = [w_t \cdot e_{nt}]_{T \times N} = [c_{nt}]_{T \times N}. \quad (25)$$

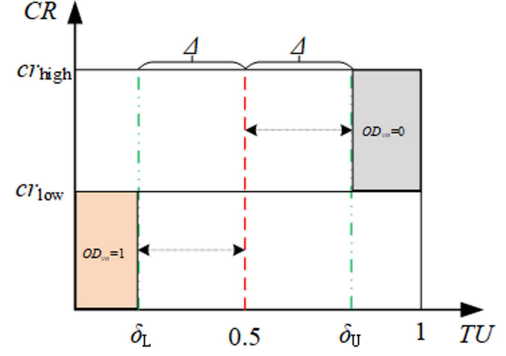


Fig. 4. Schematic diagram of bivariate truncation task triage.

where $w_t = \frac{1-E_t}{t - \sum_{t=1}^T E_t}$, $E_t = \sum_{n=1}^N \frac{P_{nt} \ln P_{nt}}{\ln N}$, $P_{nt} = o_{nt} / \sum_{n=1}^N o_{nt}$.

Step 3: Determine the ideal solution c_t^+ and the negative ideal solution c_t^- as follows:

$$\begin{cases} c_t^+ = \max \{c_{1t}, c_{2t}, \dots, c_{nt}\} \\ c_t^- = \min \{c_{1t}, c_{2t}, \dots, c_{nt}\} \end{cases}. \quad (26)$$

Step 4: The TOPSIS value, denoted as TU_n , is calculated using (27). And the comprehensive ranking of different task urgencies is determined based on this evaluation value, i.e., the larger the TU_n value, the higher the urgency.

$$TU_n = \frac{D_n^-}{D_n^+ + D_n^-}, \quad n = 1, 2, \dots, N. \quad (27)$$

where $D_n^+ = \sqrt{\sum_{t=1}^T (c_{nt} - c_t^+)^2}$, $D_n^- = \sqrt{\sum_{t=1}^T (c_{nt} - c_t^-)^2}$.

We employ a bivariate truncation method to process the task urgency dataset TU , so as to more effectively select the urgent tasks that meet the requirements. The upper truncation point $\delta_U = (0.5 + \Delta)$ and the lower truncation point $\delta_L = (0.5 - \Delta)$ are defined, where Δ represents the truncation parameter used for determining the range of task urgency data. Additionally, we conducted a comprehensive analysis of the TU and task criticality dataset CR to identify tasks with extreme characteristics, as shown in Fig. 4. The task triage can be calculated as follows:

$$\begin{cases} T_{vn}^1 = (\delta_U < TU < 1) \cup (CR = cr_{High}) \\ T_{vn}^2 = (0 < TU < \delta_L) \cup (CR = cr_{Low}) \end{cases} \quad (28)$$

For tasks that satisfy (28), the offloading decision rule is as follows:

$$OD_{vn} = \begin{cases} 0, & \text{if } T_{vn}^1 = \text{True} \\ 1, & \text{if } T_{vn}^2 = \text{True} \end{cases}. \quad (29)$$

The decision-making process for offloading is provided by Algorithm 1, which is based on a task triage strategy.

B. Proposed BME-Based MOO Algorithm

Next, we propose a MOHGA based on the BME-based MOOF to make offloading decisions for T_{remain} . While the convergence performance of the genetic algorithm of BME-based MOOF is satisfactory, there is still potential for improvement in the diversity of the solution set. The primary factor contributing to

Algorithm 1: Offloading Decision-Making Algorithm Considering Task Urgency and Criticality.

Input: Vehicle set $V = \{1, 2, \dots, V\}$; Task set T_v , with δ_U and δ_L as threshold values.

Output: 1) The task set T_{pre} , which has been preprocessed strategy, along with OD_{vn} ;

2) The set of tasks T_{remain} .

Begin

Initialize T_{pre} , T_{remain} , and OD_{vn} as empty sets.

for $v = 1$ to V **do**

for $n = 1$ to N **do**

 Compute the urgency evaluation value TU_n of task t_{vn} (using the entropy weight TOPSIS method).

 Retrieve the critical level cr_{vn} of task t_{vn} (either cr_{high} or cr_{low}).

if $\delta_U < TU_n < 1$ and $cr_{vn} == cr_{high}$ **then**

$OD_{vn}[t_{vn}] = 0$

$T_{pre} \leftarrow T_{pre} \cup \{t_{vn}\}$

else if $0 < TU_n < \delta_L$ and $cr_{vn} == cr_{low}$ **then**

$OD_{vn}[t_{vn}] = 1$

$T_{pre} \leftarrow T_{pre} \cup \{t_{vn}\}$

else

$T_{remain} \leftarrow T_{remain} \cup \{t\}$

end if

end for

end for

End

this phenomenon is the non-uniform distribution of reference points. Thus, the BME method tends to overly explore regions bounded by densely clustered reference points in the objective space, while neglecting regions with sparser reference points. Therefore, we adopt an ALIM strategy. The proposed MOO algorithm is presented in Fig. 5, where P_0 , P , P_1 , P_2 , P_3 , P_4 , and P_{new} are populations during the execution process. The population P is divided into two equal sub-populations, denoted P_1 and P_2 , in accordance with the ascending order of fitness value C_d . The better sub-population, P_1 , is selected as the parent population in the genetic phase to generate a new subpopulation, denoted as P_3 . P_3 is combined with P_2 for the associative learning phase.

1) *Population Initialization and Encoding:* The initialization of the population plays a crucial role in enhancing the search efficiency of the algorithm and improving the quality of the solution. To effectively tackle the high-dimensional characteristics of the decision space and alleviate the challenges posed by dimensionality, we employ a structured population encoding method. In this structure, each individual in the population P_0 represents a candidate solution and embodies the task offloading strategy of each CAV through a multi-layer binary vector. Each individual consists of V binary sub-vectors, where V represents the number of CAVs. The binary bits in each sub-vector are randomly generated to represent the task offloading decision of each CAV. The number of binary elements in each sub-vector's

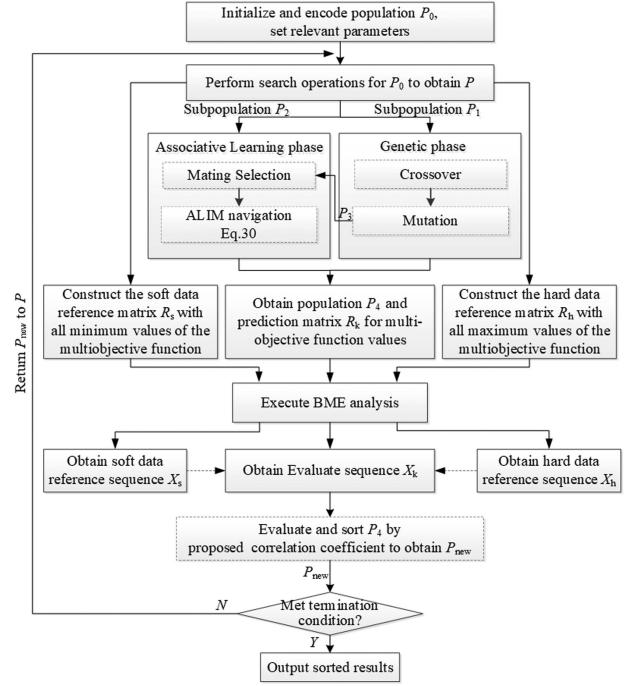


Fig. 5. BME-based MOO algorithm.

structure corresponds to T_{remain} , ensuring consistent and interpretable encoding.

2) *Fitness Evaluation With BME:* A fitness assignment is needed for each encoded solution in population P . Based on the proposed BME-based MOOF method in Section V, the fitness assignment process is presented as follows.

a) Each solution in P is decoded, and the objective function values of each solution are calculated using (7)–(9). Subsequently, the G sets of reference sequences X_h and X_s are obtained.

b) The BME method employs X_h and X_s to infer the position of each solution X_{kl} in the space. Subsequently, a spatial correlation analysis is conducted to assign a C_d value as the fitness score for each solution X_{kl} . Finally, all solutions in population P_4 are evaluated and sorted based on their respective C_d values.

3) *Genetic Operation: Step 1:* Selection operator. The binary tournament selection method is employed to choose individuals for crossover and mutation from the parent population P_1 . The specific steps are as follows:

- a) Establish an empty set to store the selected solution individuals. Prioritize individuals with smaller C_d values when selecting two individuals from the population for comparison. If C_d values are equal, prioritize individuals with a higher crowded distance.
- b) Repeat step (a) iteratively. If the newly selected individual coincides with any individual already in the set, continue re-selecting until the desired number of individuals is achieved.

Step 2: Crossover operator and mutation operator. Two different types of crossover operators are used for CAV task sequences, depending on the number of tasks. The Single-point crossover operator is utilized when the count of CAV tasks is less than

3, while the multi-point crossover operator is applied when there are 3 or more tasks. If the crossover probability is met, the crossover operation is performed on two randomly selected parents from the set that has undergone the selection operation. The single-point mutation method is employed for the mutation operation, with each solution individual being mutated after crossover with a fixed probability. A new population, denoted as P_3 , is obtained after undergoing crossover and mutation.

4) *ALIM Strategy*: The ALIM strategy initially conducts mating selection by pairing individuals from P_3 with individuals from P_2 to incorporate excellent individuals into the new population. Then, the interaction of individual learning and social learning, as shown in (30) [30], achieves a balance between local search and global search, thereby enhancing the diversity of the solution set. Finally, to form the elite population, the supplementary population obtained from ALIM is merged with population P_3 , denoted as P_4 . We employ the BME-based MOO method to evaluate P_4 . During this process, we select NP excellent solutions from the population to form a new population, denoted as P_{new} .

$$\begin{aligned} X_g^{gen+1} = & X_g^{gen} + 0.05(1 - S_1) \\ & \cdot \text{Gaussian}[X_g^{gen} - X_g^{\min}, X_g^{\max} - X_g^{gen}] \\ & + S_2 \cdot r_1(\text{Best}p^{gen} - X_g^{gen}) \\ & + S_1 \cdot r_2(\text{Best}g^{gen} - X_g^{gen}). \end{aligned} \quad (30)$$

where, gen denotes to the generation number, $G_{\max}$ represents the maximum generations number, *Gaussian* is the Gaussian distribution, r_1 and r_2 denote to the random number within $[0,1]$, *Bestp* is a guidance vector identified by the probability distribution, *Bestg* is the global solution, $S_2 = 2 - 2gen/G_{\max}$ is the cognitive factor, and $S_1 = gen/G_{\max}$ is the social factor, and $g \in (1, 2, \dots, NP)$.

C. Complexity Analysis

Our proposed collaborative task offloading approach is implemented in two stages. The first stage uses the entropy-weighted TOPSIS method with a complexity of $O(N \times T)$, which is mainly caused by data normalization and the calculation of the weighted normalization matrix. The complexity of MOHGA in the second stage is primarily due to the computation and analysis of the covariance matrix and conditional covariance matrix, which include hard data, soft data, and prediction points. Therefore, its complexity can be expressed as $O(G^2 \times NP \times M)$, whereas other computational operations exhibit lower complexity. In this paper, N is less than $G^2 \times NP$, $T = M$. As a result, the complexity of the approach is $O(G^2 \times NP \times M \times G_{\max})$.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experiment Setup

In this section, we implement the simulation model based on the system model presented in Section III. The real-world map is based on the traffic congestion area around Wuhan University of Technology, Wuhan, China. For the behavior of vehicles, we use

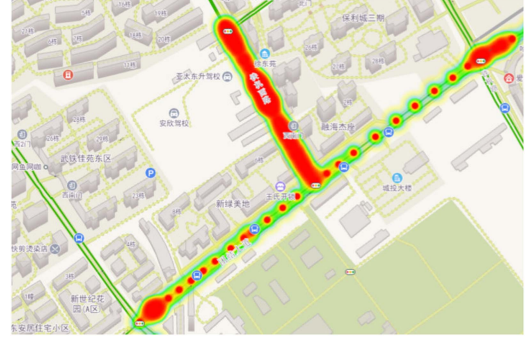


Fig. 6. Illustration of data selection in Wuhan, China.

TABLE III
PARAMETER SETTINGS

PARAMETER	VALUE
V	{50, 70, 90, 120, 150}
N	{10, 15, 20, 25}
ω_{vn}	[0.1, 1] MB
c_v^C	2GHz
C_E	20GHz
λ_v	[0.5, 1.5] tasks/s
η_v	[0.008, 0.01] CPU cycles/bit
τ_{vn}	[1, 3] s
p_v^T	[20, 30] dBm
δ^2	-100dBm
κ_v	10^{-27}
p_v^N	[0.1, 0.4]
B_v	[10, 50] MHz
π_v^E, π_v^b	1×10^{-9} USD, 1.75×10^{-9} USD
rew_{vn}	0.5 USD

part of the Amap open dataset containing mobility traces of taxi in Wuhan [31]. The illustration of the selected traffic intersection and its infrastructure location in the real-world map is depicted in Fig. 6. Based on the above information, we utilized five different scales for the vehicles, with the number of vehicles set to 50, 70, 90, 120, and 150. Additionally, the number of tasks for each CAV is set to 10, 15, 20, and 25, while the number of MRSU has been set at 4 and 6. Other parameters relevant to the experiments are shown in Table III. Similar parameter settings are also adopted in these existing literatures [6], [29]. The simulation experiment environment has been developed in MATLAB R2019b running on a Windows 10 Home Edition work station equipped with an RTX 2040 GPU and an Intel Core i7-10875H processor.

To demonstrate the superiority of our proposed method, we compare it with the following benchmark algorithm.

- *RWGA* [32]: A multi-objective genetic algorithm (MOGA) based on random weighting (RWGA), which enhances population diversity but may not accurately reflect the importance of each objective.
- *GEPAGA* [33]: A MOGA based on grey entropy parallel analysis (GEPAGA) for the fitness evaluation of solutions.
- *FREGA* [34]: A MOGA based on fuzzy relative entropy (FREGA), in which the choice of fuzzy membership functions and parameters can significantly influence its performance.

TABLE IV
MEAN HV RESULTS FOR THE FOUR MOOFS BASED ON DIFFERENT FRAMEWORKS

Inst.	Size (V×N×C)	HV			
		RWGA	GEPAGA	FREGA	MOHGA
1	50×10×4	1.0439(-)	1.0433(-)	1.0123(-)	1.1824
2	70×10×4	1.1958(-)	1.1461(-)	1.2589(-)	1.4018
3	90×10×4	1.0130(-)	0.9480(-)	1.0623(-)	1.1488
4	120×10×4	0.9010(-)	1.0360(-)	0.7920(-)	1.2822
5	120×15×4	0.8891(-)	1.0158(-)	0.7857(-)	1.2482
6	120×20×4	1.1094(-)	1.0203(-)	1.1620(-)	1.2349
7	120×20×6	1.1118(-)	1.0533(-)	1.0483(-)	1.3331
8	120×25×6	0.8959(-)	0.9494(-)	0.9087(-)	1.1657
Ave.		1.0352	1.0333	1.0147	1.2554

TABLE V
MEAN SP RESULTS FOR THE FOUR MOOFS BASED ON DIFFERENT FRAMEWORKS

Inst.	Size (V×N×C)	SP			
		RWGA	GEPAGA	FREGA	MOHGA
1	50×10×4	1.0421(-)	0.9570(-)	0.9688(-)	0.7637
2	70×10×4	1.4298(-)	1.3810(-)	1.5015(-)	0.9883
3	90×10×4	1.0062(-)	0.9584(-)	1.1079(-)	0.7432
4	120×10×4	0.9903(-)	1.0519(-)	0.8583(+)	0.9214
5	120×15×4	1.0190(-)	0.9598(+)	1.0533(-)	0.9743
6	120×20×4	1.0100(-)	0.9797(-)	1.0216(-)	0.8406
7	120×20×6	1.0171(-)	1.0433(-)	1.2078(-)	0.8108
8	150×25×6	0.9972(-)	1.0010(-)	1.0176(-)	0.8806
Ave.		1.0422	1.0057	1.0756	0.8685

- *NSGA-III* [13]: A well-known MOO algorithm commonly used for task offloading, in which a reference-point strategy is adopted to maintain population diversity.
- *MOEA/D* [14]: Task offloading algorithm based on MOEA/D in edge-assisted internet of connected vehicles, which decomposes a MOOP into multiple scalar optimization subproblems and solves them simultaneously.
- *SPEA-II* [15]: A popular multi-objective optimization evolutionary algorithm based on the improved Strength Pareto Evolutionary Algorithm (SPEA-II), which is widely used in task offloading.

For a fair comparison, we uniformly set the population size of all algorithms to 80, the crossover probability p_c to 0.9, the truncation points variable $\Delta = 0.15$, and the mutation probability p_m to 0.1 with 60 iterations per run. Each algorithm was independently run 20 times to obtain the average result for the four objectives. Two well-known metrics, namely the hypervolume (HV) and spacing (SP), were adopted to evaluate the performance of the algorithms being compared, as depicted in (31) and (32) [35], [36], respectively.

$$HV(Non, R) = \cup_{o=1}^{Non} SV(o, R). \quad (31)$$

$$SP(Non) = \sqrt{\frac{1}{|Non| - 1} \sum_{o=1}^{|Non|} (\bar{n} - n_o)^2}. \quad (32)$$

where Non denotes Pareto optimal solutions, and $|Non|$ represents the number of all optimal solutions. SV denotes the hypervolume composed of o and reference points R . n_o denotes the minimum distance from the o^{th} solution to the other solutions. $\bar{n}$ is the average of all n_o . A larger HV value indicates better convergence and diversity performance of the algorithms, while a smaller SP value indicates a better distribution of the algorithm solutions.

B. Effectiveness of MOOF Based on BME

To validate the effectiveness of the BME-based MOOF proposed in this study, we compared its performance with three other GA algorithms based on different MOOFS for solving MOTO. The only difference in the aforementioned algorithms is their MOOF. Tables IV and V present the comparison results based on the mean HV and SP, respectively. The symbols “=,”

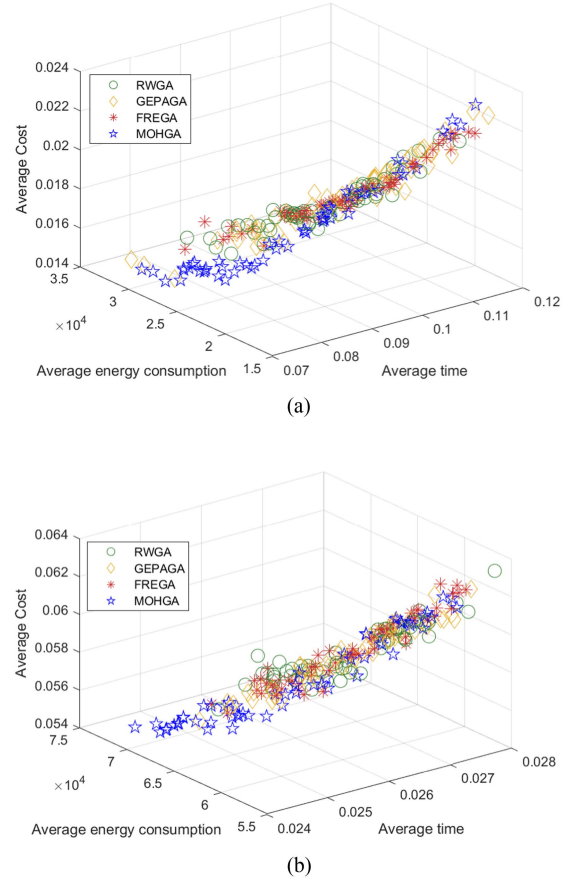
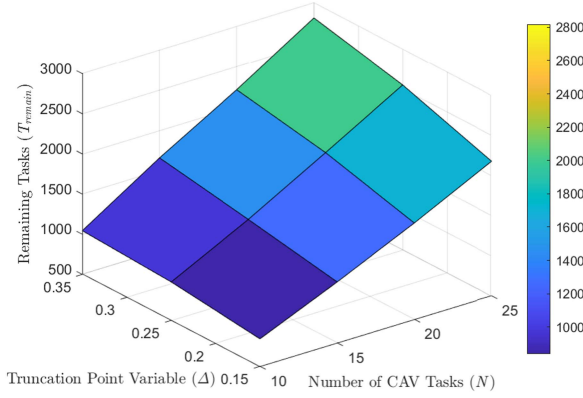


Fig. 7. Pareto fronts obtained by the four MOOFS on instance (a) 1 and instance (b) 8.

“-,” and “+” denote that the solutions are equal, worse, and better compared with MOHGA, respectively. The best results of metrics are highlighted in bold. Table IV shows that MOHGA achieves the best HV value in 8 instances, and it outperforms its competitors in all instances. From the mean values of SP in Table V, it is evident that MOHGA achieves the best SP results in 6 out of the 8 instances, whereas both GEPAGA and FREGA only achieve the best SP result once. These experimental results are further evidence of the effectiveness of BME-based MOOF in addressing the MOTO problem. Fig. 7 presents the Pareto fronts of the four algorithms on a selected medium-scale instance

Fig. 8. Relationship T_{remain} and Δ .

(1) and a large-scale instance (8). From Fig. 7, we observe that the solution set of MOHGA is closer to the origin of the coordinate axis, which intuitively illustrates that the MOHGA performs better than the other three MOOFs. Therefore, it can be stated that the proposed algorithm provides the solution set closest to the true Pareto front.

C. Effectiveness of Task Triage Strategy

Fig. 8 depicts the correlation between the truncation point variable Δ and the remaining tasks, denoted as T_{remain} , with a total of 120 CAVs. T_{remain} represents the number of remaining tasks for all tasks after the first stage of triage selection. From the figure, it is evident that as Δ increases, the remaining task T_{remain} decreases. This implies that a larger Δ may more effectively offload tasks and improve scheduling performance. However, it is crucial to acknowledge that an excessively large Δ may lead to an unreasonable allocation of certain tasks. Therefore, it is necessary to consider the balance between task distribution uniformity and enhancement of algorithmic scheduling efficiency when choosing Δ .

To further evaluate the effectiveness of the triage strategy, we conducted a comparative analysis of the average CPU time and standard deviation for MOHGA with and without the triage strategy across different task sizes, as depicted in Fig. 9. The number of CAVs is still set to 120. It can be intuitively observed from Fig. 9 that the average CPU time of the MOHGA algorithm with the triage strategy is significantly better than that without it, thus resulting in a time savings of at least 16.06%. The results indicate that the triage strategy has a notable impact on enhancing the efficiency of the algorithm.

D. Effectiveness of the ALIM Strategy

To evaluate the effectiveness of the ALIM strategy, we conducted a comparison between MOHGA1 and MOHGA2, which represent the proposed MOHGA without and with ALIM, respectively. Table V presents the mean HV and SP results obtained by the two MOHGAs. We see from Table VI that MOHGA2 performs better in all instances based on HV. In terms of SP, MOHGA2 performed better in only 6 out of the 8 instances.

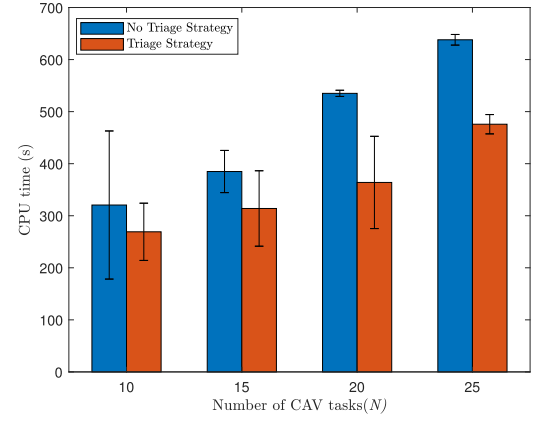


Fig. 9. Comparison of CPU time with and without triage strategy.

TABLE VI
MEAN HV AND SP RESULTS OBTAINED BY THE TWO MOHGAS WITHOUT AND WITH ALIM

Inst.	Size (V×N×C)	HV		SP	
		MOHGA1	MOHGA2	MOHGA1	MOHGA2
1	50×10×4	1.0057(-)	1.3664	1.0132(+)	1.0461
2	70×10×4	1.0392(-)	1.1251	1.0251(-)	0.8237
3	90×10×4	1.0282(-)	1.2531	0.9220(-)	0.9350
4	120×10×4	0.9248(-)	0.9908	0.9545(-)	0.8681
5	120×15×4	0.6729(-)	0.9317	1.1293(-)	0.9943
6	120×20×4	0.9683(-)	1.2765	0.9987(+)	1.0090
7	120×20×6	0.8391(-)	1.0218	1.2159(-)	0.9328
8	150×25×6	1.0316(-)	1.1420	1.0874(-)	0.9645
Ave.		0.9387	1.1384	1.0487	0.9413

Overall, the results in Table V indicate that the convergence and diversity of MOHGA2 are significantly better than those of MOHGA1. This may be attributed to the implementation of the ALIM strategy, which enhances population diversity, and the BME method, which evaluates and selects solutions more efficiently. The Pareto fronts of MOHGA1 and MOHGA2 on instances (3) and (5), as depicted in Fig. 10, show that MOHGA2 is significantly closer to the origin of the coordinate axis.

E. Comparison With Existing Algorithms

To evaluate the performance of MOHGA, we compare it with three well-known multi-objective algorithms, namely the MOEA/D, SPEA-II and NSGA-III. The reasons for selecting these three algorithms for comparisons are as follows. First, these are three popular algorithms which are usually used as baseline algorithms to tackle multi-objective task offloading. Secondly, the effectiveness of MOHGA is assessed by comparing algorithms with different strategies. Specifically, MOEA/D utilizes a decomposition strategy, SPEA-II employs an external archive, and NSGA-III adopts a reference point strategy. Tables VII and VIII show the mean HV and SP results obtained by four algorithms, respectively. From the Table VII, we can observe that MOHGA performs better than the other three algorithms on all instances. From the Table VIII, we see that the MOHGA is better than MOEA/D, SPEA-II and NSGA-III. On most instances, the MOHGA has at least

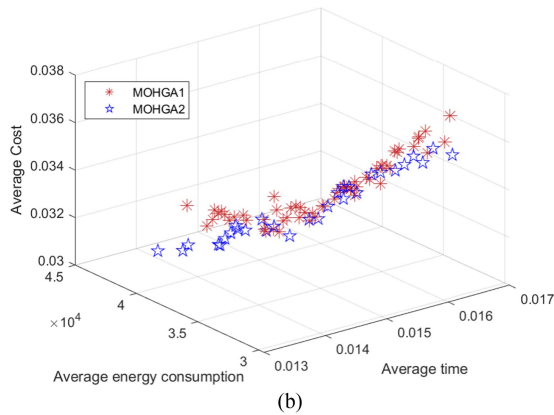
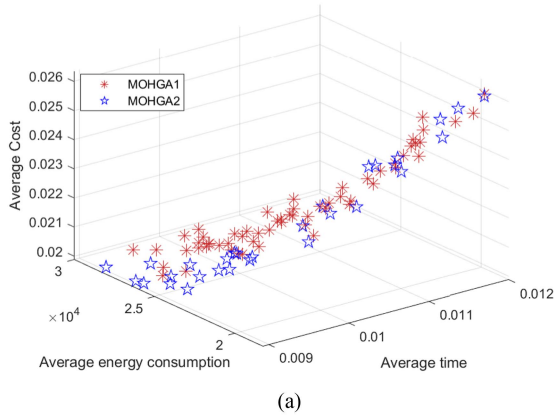


Fig. 10. Pareto fronts obtained by the two MOHGAs on instance (a) 3 and instance (b) 5.

TABLE VII
MEAN HV RESULTS OF MOEA/D, SPEA-II, NSGA-III AND MOHGA

Inst.	Size (V×N×C)	HV			
		MOEA/D	SPEA-II	NSGA-III	MOHGA
1	50×10×4	0.6814(-)	0.3516(-)	0.9542(-)	1.2212
2	70×10×4	0.7527(-)	0.4985(-)	0.8946(-)	1.0371
3	90×10×4	0.8544(-)	0.3834(-)	0.9885(-)	1.0491
4	120×10×4	0.6219(-)	0.5705(-)	0.5954(-)	0.9214
5	120×15×4	0.8492(-)	0.7476(-)	1.0084(-)	1.1478
6	120×20×4	0.7450(-)	0.5038(-)	0.8852(-)	0.9445
7	120×20×6	0.8447(-)	0.6042(-)	0.6721(-)	0.9039
8	150×25×6	0.5593(-)	0.8967(-)	0.6959(-)	1.0241
Ave.		0.7386	0.5695	0.8368	1.0311

one solution that dominates all of the solutions obtained by MOEA/D, SPEA-II and NSGA-III.

Finally, we carried out the Kruskal-Wallis ANOVA test with a significance level of 0.05 to statistically ascertain the significant dominance of the HMOGA. As can be seen in Fig. 11, the p-values of HV and SP are less than 0.05, and therefore, we can conclude that the MOHGA has significantly outperformed the other algorithms from a statistical perspective. The significant differences among these algorithms may arise from variations in key components, such as evaluation mechanisms and search operations, as well as their distinct adaptability in addressing diverse problems.

TABLE VIII
MEAN SP RESULTS OF MOEA/D, SPEA-II, NSGA-III AND MOHGA

Inst.	Size (V×N×C)	SP			
		MOEA/D	SPEA-II	NSGA-III	MOHGA
1	50×10×4	1.0554(-)	1.0558(-)	0.8191(+)	0.9941
2	70×10×4	1.0197(-)	1.0405(-)	1.0102(-)	0.9730
3	90×10×4	0.9988(-)	1.4781(-)	1.0053(-)	0.8506
4	120×10×4	1.0697(-)	0.8468(+)	0.9881(-)	0.9450
5	120×15×4	0.9644(-)	0.9211(+)	1.1822(-)	0.9329
6	120×20×4	0.9812(-)	0.9707(-)	1.0353(-)	0.9239
7	120×20×6	0.9870(-)	1.0012(-)	1.0857(-)	0.8619
8	150×25×6	0.9834(-)	0.9513(+)	1.0255(-)	0.7861
Ave.		1.0075	0.9829	1.0485	0.9291

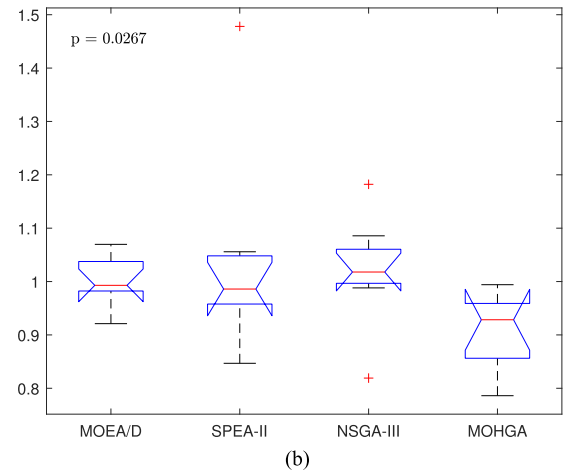
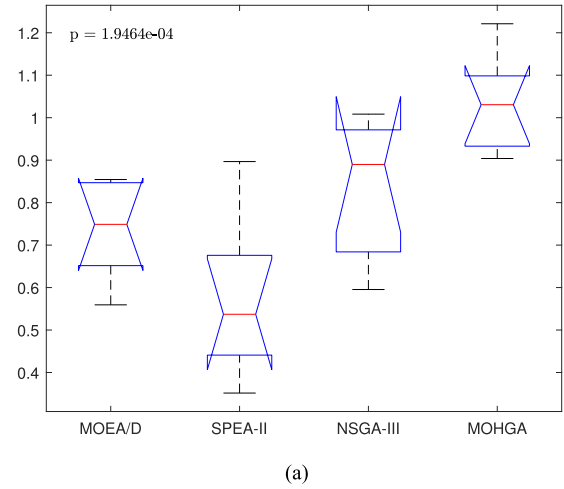


Fig. 11. Kruskal-Wallis ANOVA significance tests for the four algorithms.

VII. CONCLUSION

In this work, we propose a vehicle-centric two-stage task offloading approach, which addresses the task offloading challenge encountered by CAVs in vehicular networks. We develop a task triage strategy by analyzing the urgency and criticality of tasks, identifying those that require offloading in the first stage. Subsequently, we formulate the vehicle-centric MOTO problem by modeling the procedure of the remaining task up-load and computation based on queuing theory, which aims

at minimizing the average delay time, energy consumption, and payment cost for each vehicle. To tackle the MOTO, a novel MOOF based on BME is developed. The proposed framework integrates the ALIM strategy and genetic algorithm to develop a new MOHGA for solving the problem at hand. We conducted extensive experiments using real traffic data from Wuhan, China. The experimental results demonstrate that the proposed approach outperforms peer methods in addressing the task offloading problem, thereby confirming its effectiveness.

In future work, we will focus on several issues worthy of further research based on this paper, such as privacy-preserving cooperative offloading, joint offloading and load balancing, and dynamic offloading with mobility awareness.

REFERENCES

- [1] Z. Ning et al., "Intelligent edge computing in Internet of Vehicles: A joint computation offloading and caching solution," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 4, pp. 2212–2225, Apr. 2021.
- [2] X. Zhang, W. Wu, Z. Zhao, J. Wang, and S. Liu, "RMDDQN-learning: Computation offloading algorithm based on dynamic adaptive multi-objective reinforcement learning in Internet of Vehicles," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11374–11388, Sep. 2023.
- [3] J. Zhang and K. B. Letaief, "Mobile edge intelligence and computing for the Internet of Vehicles," *Proc. IEEE*, vol. 108, no. 2, pp. 246–261, Feb. 2020.
- [4] P. Dai, K. Hu, X. Wu, H. Xing, F. Teng, and Z. Yu, "A probabilistic approach for cooperative computation offloading in MEC-assisted vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 2, pp. 899–911, Feb. 2022.
- [5] J. Liu, J. Wan, B. Zeng, Q. Wang, H. Song, and M. Qiu, "A scalable and quick-response software defined vehicular network assisted by mobile edge computing," *IEEE Commun. Mag.*, vol. 55, no. 7, pp. 94–100, Jul. 2017.
- [6] J. Zhang, B. Gong, M. Waqas, S. Tu, and Z. Han, "A hybrid many-objective optimization algorithm for task offloading and resource allocation in multi-server mobile edge computing networks," *IEEE Trans. Serv. Comput.*, vol. 16, no. 5, pp. 3101–3114, Sep./Oct. 2023.
- [7] H. Liao et al., "Cloud-edge-Device collaborative reliable and communication-efficient digital twin for low-carbon electrical equipment management," *IEEE Trans. Ind. Inform.*, vol. 19, no. 2, pp. 1715–1724, Feb. 2023.
- [8] Y. Gao, Z. Yan, K. Zhao, T. d. Cola, and W. Li, "Joint optimization of server and service selection in satellite-terrestrial integrated edge computing networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2740–2754, Feb. 2024.
- [9] K. Peng, H. Huang, B. Zhao, A. Jolfaei, X. Xu, and M. Bilal, "Intelligent computation offloading and resource allocation in IIoT with end-edge-cloud computing using NSGA-III," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 5, pp. 3032–3046, Sep./Oct. 2023.
- [10] G. Chen, N. Guan, B. Hu, and W. Yi, "EDF-VD scheduling of flexible mixed-criticality system with multiple-shot transitions," *IEEE Trans. Comput. Aided Des. Integr. Circuits Syst.*, vol. 37, no. 11, pp. 2393–2403, Nov. 2018.
- [11] S. Safari, H. Khdr, P. Gohari-Nazari, M. Ansari, S. Hessabi, and J. Henkel, "TherMa-MiCs: Thermal-aware scheduling for fault-tolerant mixed-criticality systems," *IEEE Trans. Parallel Distrib. Syst.*, vol. 33, no. 7, pp. 1678–1694, Jul. 2022.
- [12] L. Cui et al., "Joint optimization of energy consumption and latency in mobile edge computing for Internet of Things," *IEEE Internet Things J.*, vol. 6, no. 3, pp. 4791–4803, Jun. 2019.
- [13] Q. Li, Z. Shi, Z. Xue, Z. Cui, and Y. Xu, "A many-objective evolutionary algorithm for solving computation offloading problems under uncertain communication conditions," *Comput. Commun.*, vol. 213, pp. 22–32, 2024.
- [14] X. Xu, X. Zhang, X. Liu, J. Jiang, L. Qi, and M. Z. A. Bhuiyan, "Adaptive computation offloading with edge for 5G-envisioned internet of connected vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 8, pp. 5213–5222, Aug. 2021.
- [15] Q. Wu, X. Xu, Q. Zhao, and F. Dai, "Tasks offloading for connected autonomous vehicles in edge computing," *Mobile Netw. Appl.*, vol. 27, no. 6, pp. 2295–2304, 2022.
- [16] X. Huang, L. He, X. Chen, L. Wang, and F. Li, "Revenue and energy efficiency-driven delay-constrained computing task offloading and resource allocation in a vehicular edge computing network: A deep reinforcement learning approach," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8852–8868, Jun. 2022.
- [17] M. Hao, D. Ye, S. Wang, B. Tan, and R. Yu, "URLLC resource slicing and scheduling for trustworthy 6G vehicular services: A federated reinforcement learning approach," *Phys. Commun.*, vol. 49, 2021, Art. no. 101470.
- [18] B. Hu, Z. Yan, and M. Zhao, "Workload-aware scheduling of multiple-criticality real-time applications in vehicular edge computing system," *IEEE Trans. Ind. Inform.*, vol. 19, no. 10, pp. 10091–10101, Oct. 2023.
- [19] X. Hou et al., "Reliable computation offloading for edge-computing-enabled software-defined IoV," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7097–7111, Aug. 2020.
- [20] Q. Shen, B.-J. Hu, and E. Xia, "Dependency-aware task offloading and service caching in vehicular edge computing," *IEEE Trans. Veh. Technol.*, vol. 71, no. 12, pp. 13182–13197, Dec. 2022.
- [21] W. Fan et al., "Game-based task offloading and resource allocation for vehicular edge computing with edge-edge cooperation," *IEEE Trans. Veh. Technol.*, vol. 72, no. 6, pp. 7857–7870, Jun. 2023.
- [22] M. Jia, J. Cao, and W. Liang, "Optimal cloudlet placement and user to cloudlet allocation in wireless metropolitan area networks," *IEEE Trans. Cloud Comput.*, vol. 5, no. 4, pp. 725–737, Oct.–Dec. 2017.
- [23] J. He and A. Kolovos, "Bayesian maximum entropy approach and its applications: A review," *Stochastic Environ. Res. Risk Assessment*, vol. 32, pp. 859–877, 2018.
- [24] J. Wang, S. Kang, J. Sun, Z. Chen, and N. Song, "Spatial prediction of crop water requirement based on Bayesian maximum entropy and multi-source data," *Trans. Chin. Soc. Agricultural Eng.*, vol. 33, no. 9, pp. 99–106, 2017.
- [25] M. L. Serre, P. Bogaert, and G. Christakos, "Computational investigations of Bayesian maximum entropy spatiotemporal mapping," in *Proc. IAMG 4th Annu. Conf. Int. Assoc. Math. Geol.*, vol. 1, 1998, pp. 117–122.
- [26] Y. Liu, J. Li, and S. Sun, "Advances in Gaussian random field generation: A review," *Comput. Geosciences*, vol. 23, pp. 1011–1047, 2019.
- [27] M. Serre and G. Christakos, "Modern geostatistics: Computational BME analysis in the light of uncertain physical knowledge—the Equus Beds study," *Stochastic Environ. Res. Risk Assessment*, vol. 13, pp. 1–26, 1999.
- [28] C. Jin-qiang, "Fault prediction of a transformer bushing based on entropy weight TOPSIS and gray theory," *Comput. Sci. Eng.*, vol. 21, no. 6, pp. 55–62, 2019.
- [29] R. Liu, T. Yang, G. Sun, S. Lin, F. Li, and X. Wang, "Multi-objective optimal scheduling of community integrated energy system considering comprehensive customer dissatisfaction model," *IEEE Trans. Power Syst.*, vol. 38, no. 5, pp. 4328–4340, Sep. 2023.
- [30] A. A. Heidari, I. Aljarah, H. Faris, H. Chen, J. Luo, and S. Mirjalili, "An enhanced associative learning-based exploratory whale optimizer for global optimization," *Neural Comput. Appl.*, vol. 32, pp. 5185–5211, 2020.
- [31] Q. Liu, B. Chen, and X. Li, "Construction of a micro model for CO2 emissions from CNG taxi based on trajectory data and deep learning method and evaluation of carbon reduction benefits," *J. Geo-inf. Sci.*, vol. 25, no. 11, pp. 2191–2203, 2023.
- [32] A. Konak, D. W. Coit, and A. E. Smith, "Multi-objective optimization using genetic algorithms: A tutorial," *Rel. Eng. System Safe.*, vol. 91, no. 9, pp. 992–1007, 2006.
- [33] L. He, R. Chiong, and W. Li, "Energy-efficient open-shop scheduling with multiple automated guided vehicles and deteriorating jobs," *J. Ind. Inform. Integr.*, vol. 30, 2022, Art. no. 100387.
- [34] G. Zhu, C. Ding, and W. Zhang, "Optimal foraging algorithm that incorporates fuzzy relative entropy for solving many-objective permutation flow shop scheduling problems," *IEEE Trans. Fuzzy Syst.*, vol. 28, no. 11, pp. 2738–2746, Nov. 2020.
- [35] Z. Cui, Z. Xue, T. Fan, X. Cai, and W. Zhang, "A many-objective evolutionary algorithm based on constraints for collaborative computation offloading," *Swarm Evol. Computation*, vol. 77, 2023, Art. no. 101244.
- [36] S. Mousavi, S. E. Mood, A. Souiri, and M. M. Javidi, "Directed search: A new operator in NSGA-II for task scheduling in IoT based on cloud-fog computing," *IEEE Trans. Cloud Comput.*, vol. 11, no. 2, pp. 2144–2157, Apr.–Jun. 2023.



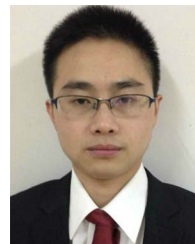
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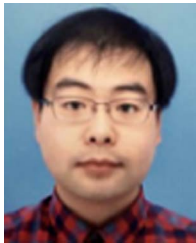
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